

1 · INPUT DATA

Bathymetry & coastline

ETOPO2 · GSHHS
→ grid, mask, smoothing $rx0 \leq 0.2$

Ocean: GLORYS12 reanalysis

daily, 1/12°, 50 levels (copernicus-marine ARCO) · Mercator anfc if recent

Atmosphere: ERA5

hourly, 0.25° (CDS ARCO Zarr)
bulk fluxes, interpolated ONLINE

Tides: TPX07 (optional)

10 constituents M2 S2 N2 K2 K1 O1
P1 Q1 Mf Mm · SSH, UV, potential

Rivers: GloFAS v4 (optional)

daily discharge (EWDS) at Dai & Trenberth mouths · $Q \geq 100 \text{ m}^3/\text{s}$

2 · BUILD a configuration — build_config.sh NAME

domain.cfg

the only hand-written file:
box · 1/RES · period · options

01 grid

grid.ini → croco_grd.nc

02 check

mask → OBC
CFL, CARD.md

03 data

GLORYS
+ ERA5

04 ini / bry

initial +
boundaries

05 sources

cppdefs.h
param.h

06 croco.in

dt, outputs,
ERA5 online

07 compile

croco_plain
[_tides][_rivers]

08 proof run

7-day test

09 tides

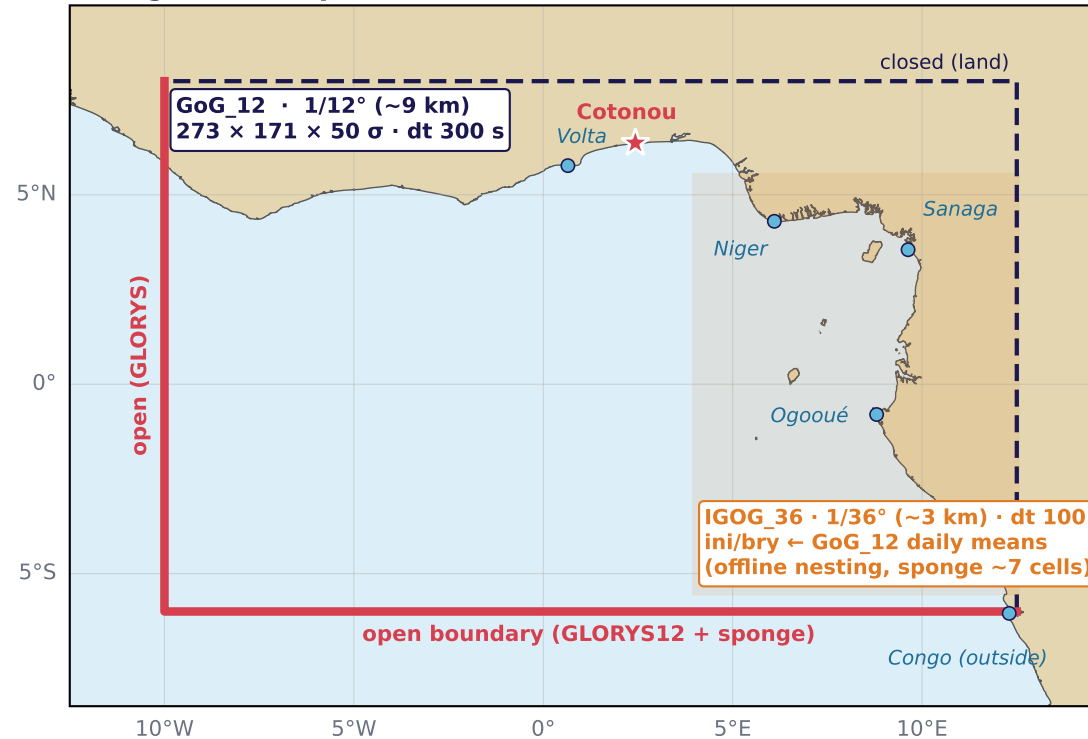
croco_frc.nc
(optional)

10 rivers

croco_runoff.nc
(optional)

each step is a script (steps/NN_*.sh), stops at the first failure; resume with --from N

configurations: parent and nested child



model set-up

Vertical grid

50 σ -levels, $\theta_s = 7$,
 $\theta_b = 2$, $hc = 200 \text{ m}$

Time stepping

$dt = 300 \text{ s}$ (1/12°),
 100 s (1/36°);
NDTFAST = 60

Stability

$CFL < 0.7$ checked
in step 02

Time origin

$Y_{orig} = 1993$

Surface forcing

ERA5 bulk formulae,
ONLINE interpolation

Outputs

his: 6-hourly (hourly
with tides) · avg: daily
· rst: daily

4 · OUTPUTS & USES

model-runs/NAME/<YYYYMMDD>_<build>/

spinup/ hcast/ gen_*/

croco_his.nc snapshots (u, v, w, T, S, ζ)
croco_avg.nc daily means
croco_rst.nc restart

log ends with **MAIN: DONE**

Validation

verify_config.py NAME: vs GLORYS
daily means, SST, tide amplitude (M2)

Offline nesting (child)

parent croco_avg.nc → nesting.py:
 $\sigma \rightarrow 50$ z-levels, rotate to E/N →
child ini/bry (daily means: no tidal
aliasing; child adds TPXO itself)

Lagrangian drift

OpenDrift reader_ROMS_native
(+ u/v masks) → oil · SAR · plastics

Forecast mode

same recipe, forcing from
CMEMS anfc + GFS / WRF

3 · RUN the cycled hindcast — run_hindcast_cycle.sh NAME

every cycle restarts from GLORYS through its own spin-up → anchored to the reanalysis, no drift; hindcast windows tile the period · bry from GLORYS ($T - 1 \rightarrow T + H + 1$) · re-runs skip finished cycles

